

# Applications of Wireless Communication

Wireless communication has dramatically changed how we talk to people and use technology. We can do things we never could with mobile phones, Wi-Fi, Bluetooth, and satellite navigation! It is essential to know how these things work and how they affect our lives as we use them more daily. There are many applications of wireless communication, such as mobile communications, the Internet of Things (IoT), wireless sensor networks, satellite communication, wireless local area networks (WLAN), and wireless power transfer (WPT).

## What is wireless communication?

Wireless communication is a way of sending information between devices without using wires. It is really popular because it is easy and convenient. There are four important things that we need for wireless communication to work:

- **Transmitter:** A transmitter is a device that sends information, like a radio or a smartphone. It's like a person who talks to you on the phone. They are the ones sending the message.
- **Receiver:** A receiver is a device that receives information, like another radio or a smartphone. It's like a person who listens to you on the phone. They are the ones getting the message.
- **Medium:** A medium is a thing that the information travels through, like air, water, or space. It's like the air that carries the sound waves from a person talking to you on the phone.
- **Signal:** A signal is an actual information that is being sent, like voice, text, video, or data. It's like the words that a person says to you on the phone.

There are different types of wireless communication that use different things to send and receive information.

- **Radio communication:** This type of wireless communication uses radio waves to send and receive information. It's like a walkie-talkie or a car radio.
- **Infrared communication:** This type of wireless communication uses light waves that we can't see to send and receive information. It's like a TV remote control or a sensor that turns on the lights when you walk into a room.
- **Bluetooth communication:** This type of wireless communication uses short-range radio waves to connect devices together. It's like when you connect your phone to a wireless speaker or a pair of headphones.

- **Wi-Fi communication:** This type of wireless communication uses radio waves to send and receive information over a longer distance than Bluetooth. It's like when you connect your computer to the internet wirelessly.

In summary, wireless communication is a way of sending information between devices without wires. It requires a transmitter, receiver, medium, and signal to work. Different types of wireless communication use different things to send and receive information. There are many applications of wireless communication, such as mobile communications, the Internet of Things (IoT), wireless sensor networks, satellite communication, wireless local area networks (WLAN), and wireless power transfer (WPT). Understanding these basics can help you better understand the technology all around us!

### **Characteristics and Features of Wireless Communication**

Wireless communication has some remarkable features that make it useful.

- It lets us communicate and send information even when moving around, called mobility. We can use our phones or other devices anywhere in the network range.
- It's convenient because we don't have to use any wires or cables to connect, which makes it easier to access information and talk to other people.
- Wireless communication is getting faster and faster, which is great for sending data quickly. Wireless networks can be made bigger to support more devices and users, called scalability.
- Security is essential in wireless communication, but there are ways to protect our information and privacy.
- Wireless communication is usually cheaper than wired networks, especially when installing wires is complex or expensive.
- Many different devices can use wireless communication, making it easier for us to connect and talk to each other.

### **Applications of Wireless Communications**

Wireless technology has changed how we talk to each other and use things daily. It lets us send and get information from almost everywhere in the world. Here are some applications of wireless communication:

- **Mobile communications:** We use things like cell phones and satellites to talk and send messages over long distances.
- **Internet of Things (IoT):** This is when we connect different gadgets over the internet, like innovative home products and industrial machines.
- **Wireless Sensor Networks (WSNs):** These tiny sensors can gather information like temperature and pressure and send it to a central system. We use them for things like traffic management and environmental monitoring.
- **Satellite communication:** We use satellites to send and get information, like weather forecasting and TV broadcasting.

- **Wireless local area networks:** This is when we connect devices to the internet without wires, like in our homes or public places like airports.
- **Wireless power transfer (WPT):** We send electrical power without cords, which we use in medical equipment and electric cars.

## Types of Telecommunications

The following list of different types of telecommunications networks includes:

1. **LAN:** Local area networks, or LANs, are made for compact spaces like offices, clusters of buildings, or factories. Since They are simple to construct and debug, LANs are frequently used. Through the LAN, the PCs and workstations are linked. LAN can be as basic as connecting two computers so they can exchange data and communicate, or it can be as sophisticated as linking a whole building.

Features:

- Resource sharing
  - Software applications sharing
  - Simple and cheap communication
  - Centralized data
  - Data security
  - Internet sharing
2. **WAN:** A wide area network, or WAN, may be rented out publicly or privately. It is used for networks that span wide areas, such as those that connect the states of a nation. This category can include networks that serve a sizable metropolis or city. Many international corporations use WANs to send and receive data among their staff, vendors, clients, and other organizations throughout many cities, regions, nations, and the whole world.

Features:

- Covers a sizable area
  - Sends messages containing data, pictures, or audio quickly
  - Connects workstations and distributes resources and applications
  - Everyone can use the same data from WANs
3. **MAN:** Also known as Metropolitan Area Network, MAN employs LAN-like technologies though it is a significantly larger variant. It is

intended to cover the entire city. It frequently entails combining many local area networks (LANs) into a single, larger network. It is mostly owned and run by one private firm or one publicly traded company.

Features:

- Efficient transmission of data via high-speed carriers.
  - Offers a single backbone for a sizable network and more WAN access
4. **Internetworks:** Internetworks are links between two or more networks. In other words, it uses a variety of hardware, including routers, bridges, and gateways, to join two or more separate networks.
  5. **Intranet and Extranet:** A telecommunication network called an intranet is intended to be open, but it is protected by internal networks whose web browser software enables users to quickly point and click their way to multimedia content on the intranet's internal internet sites. Whereas extranets are also telecommunication networks that link intranet resources of a company with other organizations and individuals. Extranets allow companies to securely communicate information over the internet by allowing authorized outside users to access portions of an intranet.
  6. **Wireless Networks:** Every other type of telecommunications network pales in comparison to wireless. It is not a new concept; in the past, wireless communications were implemented using morse code. The idea behind this system is unchanged despite the fact that digital wireless network systems now have far superior systems.

#### Telecommunication Network Examples

Telecommunication networks include the following:

- Telephones, microwave communication systems, fiber optics, telegraph, radio, satellites, and the Internet are all examples of telecommunications network equipment.
- While most people think of telecommunications with current technology, it was also utilized in ancient times to send smoke signals as a type of visual telegraph. Smoke signals, which have been widely utilized in the past, are used to send brief messages across great distances, assuming a clear line of sight.
- Cell phones with conference calling capabilities are also included in telecommunications.
- Astronauts on the Moon were able to interact with people on Earth because of advances in telecommunication technology.

- Telecommunication satellites are employed for this purpose. Satellite phones come in a variety of sizes, but they all rely on a satellite network.
- Telecommunications are carried out via cell phone towers.
- A GSM phone is a type of telecommunications equipment.
- Telecommunication networks enable command and control systems to control military troops from a remote headquarters.
- A heliograph is an optical telegraph that mimics a sign lamp by using a mirror to reflect light.

### **Different standards of Wi-Fi :**

These are the Wi-Fi standards that evolved from 1997 to 2021. In 1997 IEEE created one standard and gave the name 802.11.

#### **IEEE 802.11 –**

1. It was developed in 1997.
2. Speed is about 2 Mbps (2 megabits per second).

#### **IEEE 802.11a –**

1. This standard is developed in 1999.
2. 802.11a is useful for commercial and industrial purposes.
3. It works on a 5 GHz frequency.
4. The maximum speed of 802.11a is 54 Mbps.
5. This standard was made to avoid interference with other devices which use the 2.4 GHz band.

#### **IEEE 802.11b –**

1. This standard also created with 802.11a in 1999.
2. The difference is that it uses a 2.4 GHz frequency band.
3. The speed of 802.11b is 11 Mbps.
4. This standard is useful for home and domestic use.

#### **IEEE 802.11g –**

1. This standard is designed in 2003.
2. Basically, it has combined the properties of both 802.11a and 802.11b.
3. The frequency band used in this is 2.4 GHz for better coverage.
4. And the maximum speed is also up to 54 Mbps.

#### **IEEE 802.11n –**

1. This was introduced in 2009.
2. 802.11n operates on both 2.4 GHz and 5 GHz frequency bands, they are operated individually.
3. The data transfer rate is around 600 Mbps.

#### **IEEE 802.11ac –**

1. This standard is developed in 2013 named 802.11ac.
2. Wi-Fi 802.11ac works on the 5 GHz band.
3. The maximum speed of this standard is 1.3 Gbps.
4. It gives less range because of the 5 GHz band, but nowadays most of the devices are working on 802.11n and 802.11ac standards.

### IEEE 802.11ax –

1. It is the newest and advanced version of Wi-Fi.
2. This is released in 2019.
3. Operates on both 2.4 GHz and 5 GHz for better coverage as well as better speed.
4. User will get 10 Gbps of maximum speed around 30-40 % improvement over 802.11ac

**Wireless LANs** are those **Local Area Networks** that use high frequency radio waves instead of cables for connecting the devices in LAN. Users connected by WLANs can move around within the area of network coverage. Most WLANs are based upon the standard IEEE 802.11 or WiFi.

### IEEE 802.11 Architecture

The components of an IEEE 802.11 architecture are as follows

**1) Stations (STA)** – Stations comprise all devices and equipments that are connected to the wireless LAN. A station can be of two types:

- **Wireless Access Pointz (WAP)** – WAPs or simply access points (AP) are generally wireless routers that form the base stations or access.
- **Client.** – Clients are workstations, computers, laptops, printers, smartphones, etc.

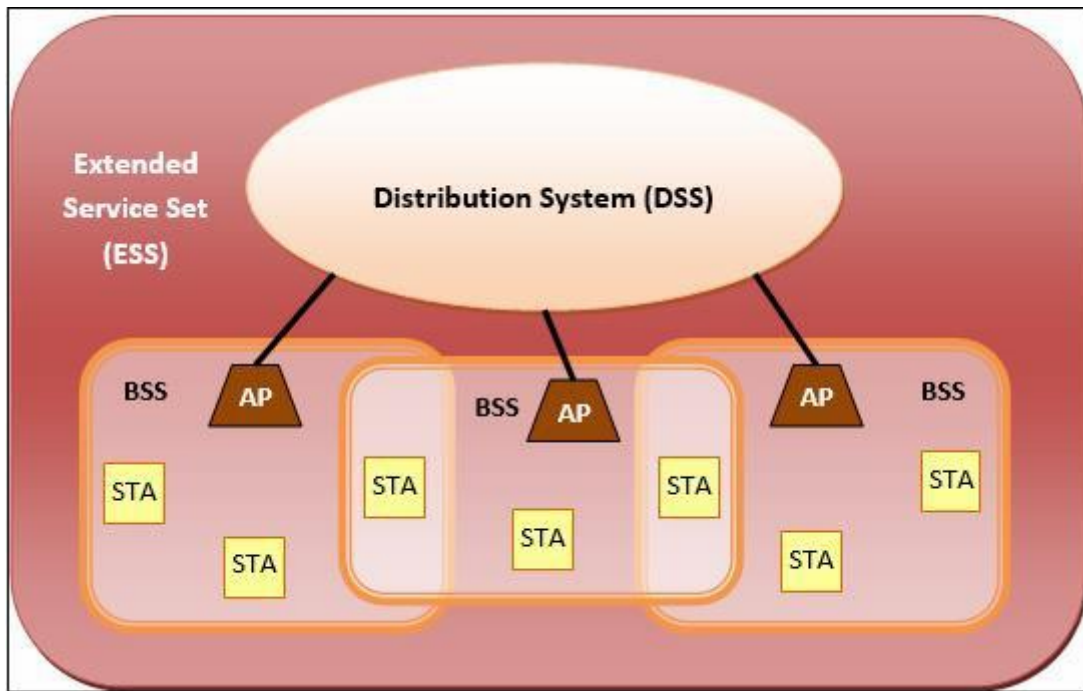
Each station has a wireless network interface controller.

**2) Basic Service Set (BSS)** –A basic service set is a group of stations communicating at physical layer level. BSS can be of two categories depending upon mode of operation:

- **Infrastructure BSS** – Here, the devices communicate with other devices through access points.
- **Independent BSS** – Here, the devices communicate in peer-to-peer basis in an ad hoc manner.

**3) Extended Service Set (ESS)** – It is a set of all connected BSS.

**4) Distribution System (DS)** – It connects access points in ESS.



### Advantages of WLANs

- They provide clutter free homes, offices and other networked places.
- The LANs are scalable in nature, i.e. devices may be added or removed from the network at a greater ease than wired LANs.
- The system is portable within the network coverage and access to the network is not bounded by the length of the cables.
- Installation and setup is much easier than wired counterparts.
- The equipment and setup costs are reduced.

### Disadvantages of WLANs

- Since radio waves are used for communications, the signals are noisier with more interference from nearby systems.
- Greater care is needed for encrypting information. Also, they are more prone to errors. So, they require greater bandwidth than the wired LANs.
- WLANs are slower than wired LANs.

## Bluetooth

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Bluetooth is universal for short-range wireless voice and data communication. It is a Wireless Personal Area Network (WPAN) technology and is used for exchanging data over smaller distances. This technology was invented by Ericson in 1994. It operates in the unlicensed, industrial, scientific, and medical (ISM) band from 2.4 GHz to 2.485 GHz. Maximum devices that can be connected at the same time are 7. Bluetooth ranges up to 10 meters. It provides data rates up to 1 Mbps or 3 Mbps depending upon the version. The

spreading technique that it uses is FHSS (Frequency-hopping spread spectrum). A Bluetooth network is called a **piconet** and a collection of interconnected piconets is called **scatternet**.

## What is Bluetooth?

Bluetooth simply follows the principle of transmitting and receiving data using radio waves. It can be paired with the other device which has also Bluetooth but it should be within the estimated communication range to connect. When two devices start to share data, they form a network called piconet which can further accommodate more than five devices.

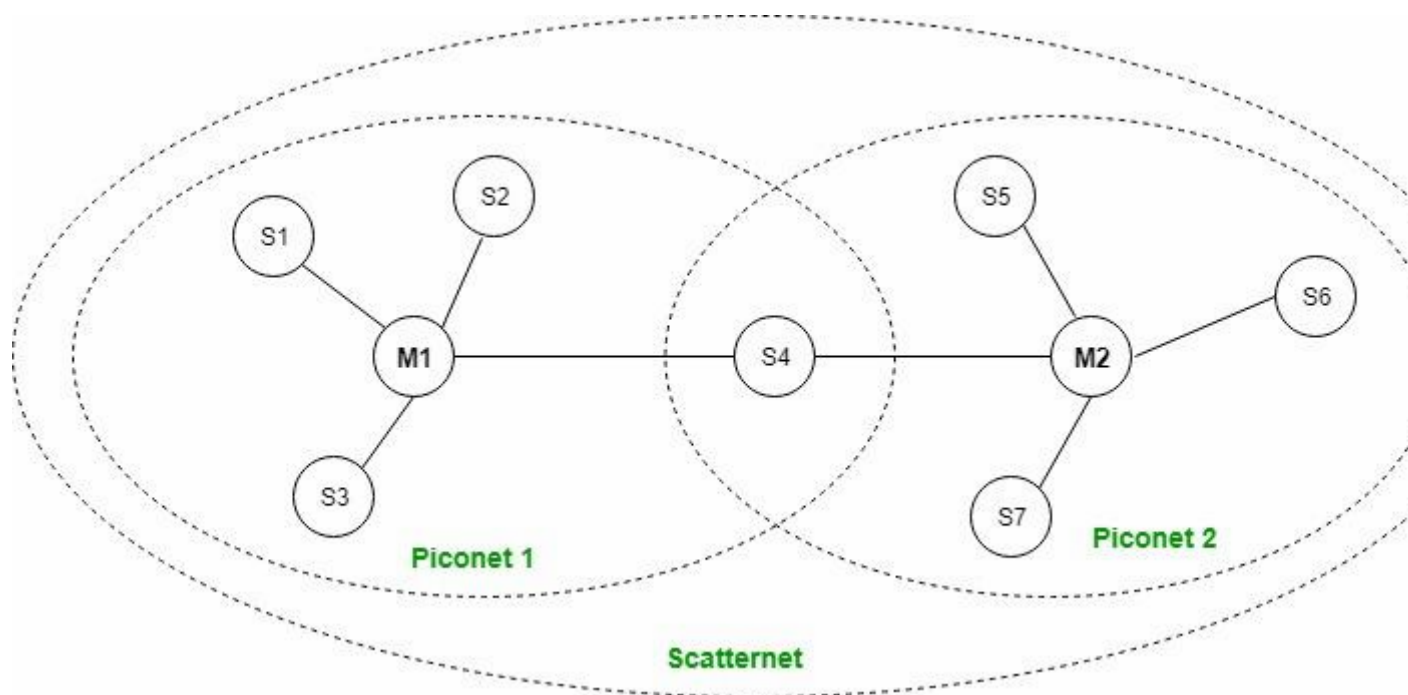
### Points to remember for Bluetooth:

- Bluetooth Transmission capacity 720 kbps.
- Bluetooth is Wireless.
- Bluetooth is a Low-cost short-distance radio communications standard.
- Bluetooth is robust and flexible.
- Bluetooth is cable replacement technology that can be used to connect almost any device to any other device.
- The basic architecture unit of Bluetooth is a piconet.

## Bluetooth Architecture:

The architecture of Bluetooth defines two types of networks:

1. Piconet
2. Scatternet



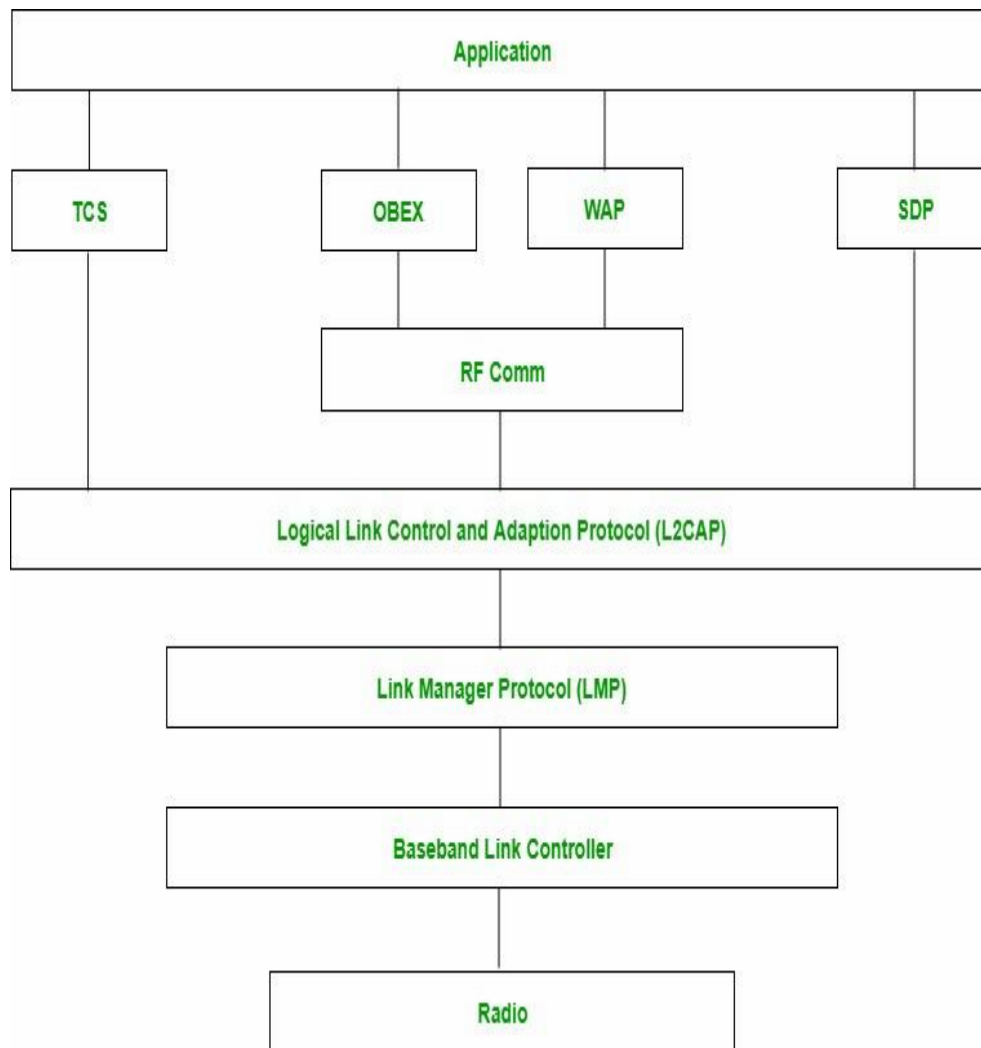
## Piconet:

Piconet is a type of Bluetooth network that contains **one primary node** called the master node and **seven active secondary nodes** called slave nodes. Thus, we can say that there is a total of 8 active nodes which are present at a distance of 10 meters. The communication between the primary and secondary nodes can be one-to-one or one-to-many. Possible communication is only between the master and slave; Slave-slave communication is not possible. It also has **255 parked nodes**, these are secondary nodes and cannot take participation in communication unless it gets converted to the active state.

## Scatternet:

It is formed by using **various piconets**. A slave that is present in one piconet can act as master or we can say primary in another piconet. This kind of node can receive a message from a master in one piconet and deliver the message to its slave in the other piconet where it is acting as a master. This type of node is referred to as a bridge node. A station cannot be mastered in two piconets.

## Bluetooth protocol stack:

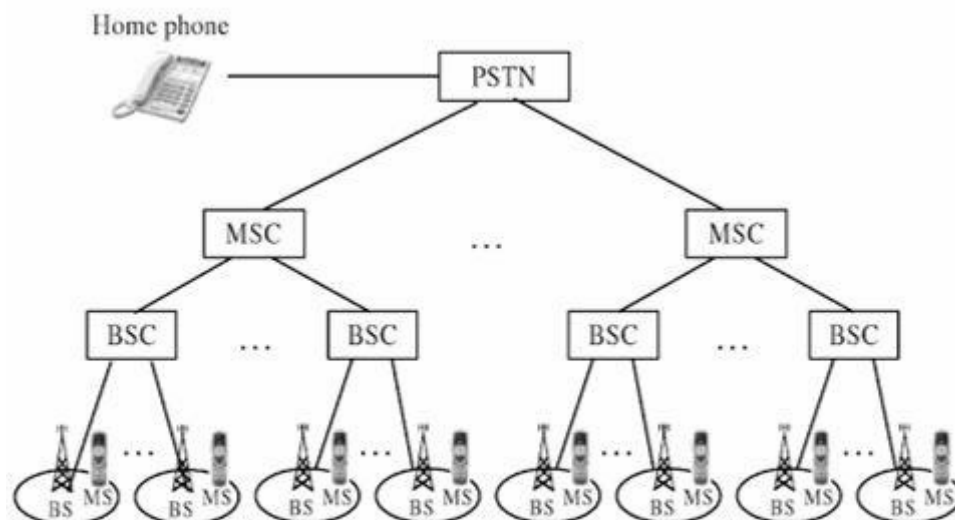


1. **Radio (RF) layer:** It specifies the details of the air interface, including frequency, the use of frequency hopping and transmit power. It performs modulation/demodulation of the data into RF signals. It defines the physical characteristics of Bluetooth transceivers. It defines two types of physical links: connection-less and connection-oriented.
2. **Baseband Link layer:** The baseband is the digital engine of a Bluetooth system and is equivalent to the MAC sublayer in LANs. It performs the connection establishment within a piconet, addressing, packet format, timing and power control.
3. **Link Manager protocol layer:** It performs the management of the already established links which includes authentication and encryption processes. It is responsible for creating the links, monitoring their health, and terminating them gracefully upon command or failure.
4. **Logical Link Control and Adaption (L2CAP) Protocol layer:** It is also known as the heart of the Bluetooth protocol stack. It allows the communication between upper and lower layers of the Bluetooth protocol stack. It packages the data packets received from upper layers into the form expected by lower layers. It also performs segmentation and multiplexing.

5. **Service Discovery Protocol (SDP) layer:** It is short for Service Discovery Protocol. It allows discovering the services available on another Bluetooth-enabled device.
6. **RF comm layer:** It is a cabal replacement protocol. It is short for Radio Frontend Component. It provides a serial interface with WAP and OBEX. It also provides emulation of serial ports over the logical link control and adaption protocol(L2CAP). The protocol is based on the ETSI standard TS 07.10.
7. **OBEX:** It is short for Object Exchange. It is a communication protocol to exchange objects between 2 devices.
8. **WAP:** It is short for Wireless Access Protocol. It is used for internet access.
9. **TCS:** It is short for Telephony Control Protocol. It provides telephony service. The basic function of this layer is call control (setup & release) and group management for the gateway serving multiple devices.
10. **Application layer:** It enables the user to interact with the application.

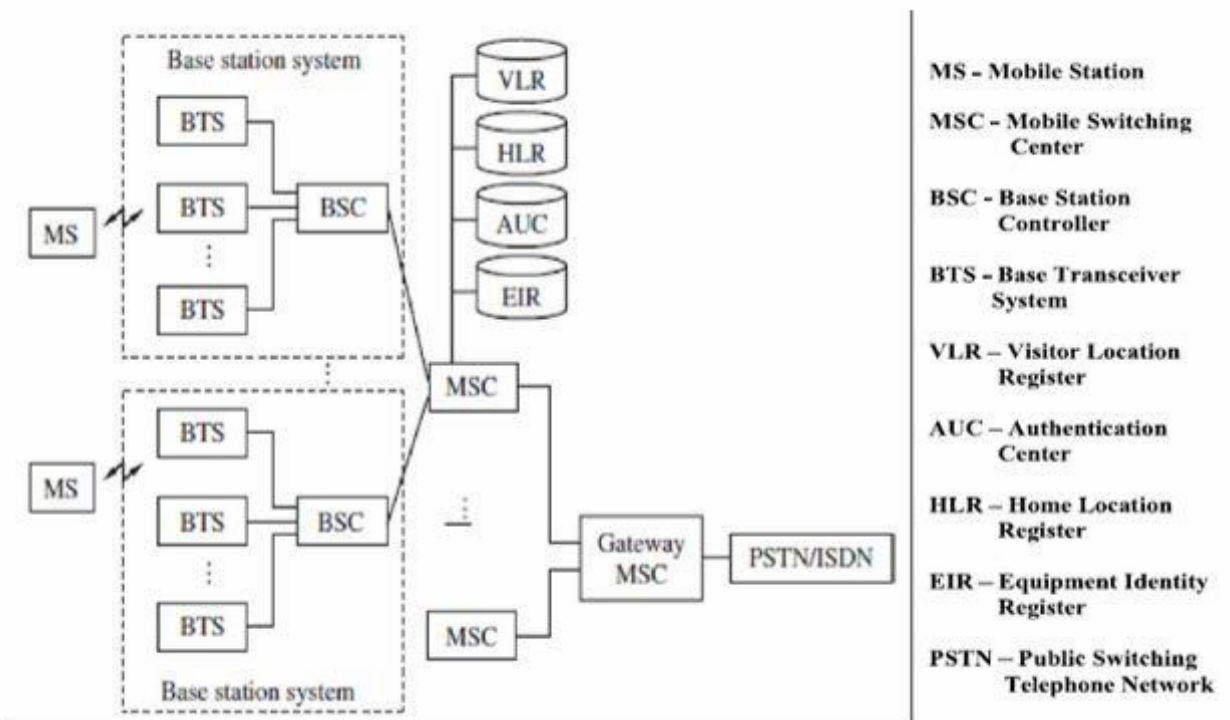
### Architecture of mobile telecommunication system

In a cellular structure, a MS (mobile station) needs to communicate with the BS of the cell where the MS is currently located and the BS acts as a gateway to the rest of the world. Therefore, to provide a link, the MS needs to be in the area of one of the cells (and hence a BS) so that mobility of the MS can be supported. Several base stations are connected through hard-wires and are controlled by a BS controller (BSC), which in turn is connected to a mobile switching center (MSC).



**Fig: cellular system infrastructure**

A cellular system requires a fairly complex infrastructure. A generic block diagram is shown in the figure:



A BS consists of a base transceiver system (BTS) and a BSC. Both tower and antenna are a part of the BTS, while all associated electronics are contained in the BSC.

The HLR (home location register) and VLR (visitor location register) are two sets of pointers that support mobility and enable the use of the same telephone numbers worldwide.

The AUC (authentication center) unit provides authentication and encryption parameters that verify the user's identity and ensure the confidentiality of each cell.

The EIR (equipment identity register) is a database that information about identity of mobile equipment. Both AUC and EIR can be implemented as individual stand-alone units or as a combined AUC/EIR unit.